3) autonomous rescue robot;

```
import heapq


# S = Start, G = Goal
# # = Blocked cell, R = Risky zone

grid = [
    ['S', '.', 'R', '#', '.'],
    ['#', '.', '.', '#', '.'],
    ['.', '.', '#', '.', '.'],
    ['.', '#', '.', 'R', '.'],
    ['.', '.', '.', '.', 'G']
]

rows = len(grid)
cols = len(grid[0])

start = (0, 0)
goal = (4, 4)

# Up, Down, Left, Right
directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]

# Priority queue: (cost, position, path)
pq = [(0, start, [start])]
visited = {}

while pq:
    cost, current, path = heapq.heappop(pq)

    if current in visited:
        continue

    visited[current] = cost

    if current == goal:
        print("Goal reached!")
        print("Path:", path)
        print("Total cost:", cost)
        break

    r, c = current
```

```
    for dr, dc in directions:
        nr = r + dr
        nc = c + dc

        if 0 <= nr < rows and 0 <= nc < cols:
            if grid[nr][nc] == '#':
                continue

            next_pos = (nr, nc)

            if next_pos not in visited:
                move_cost = 1

                if grid[nr][nc] == 'R':
                    move_cost += 2


                heapq.heappush(
                    pq,
                    (cost + move_cost,
                     next_pos,
                     path + [next_pos])
                )
else:
    print("No safe path found!")
```